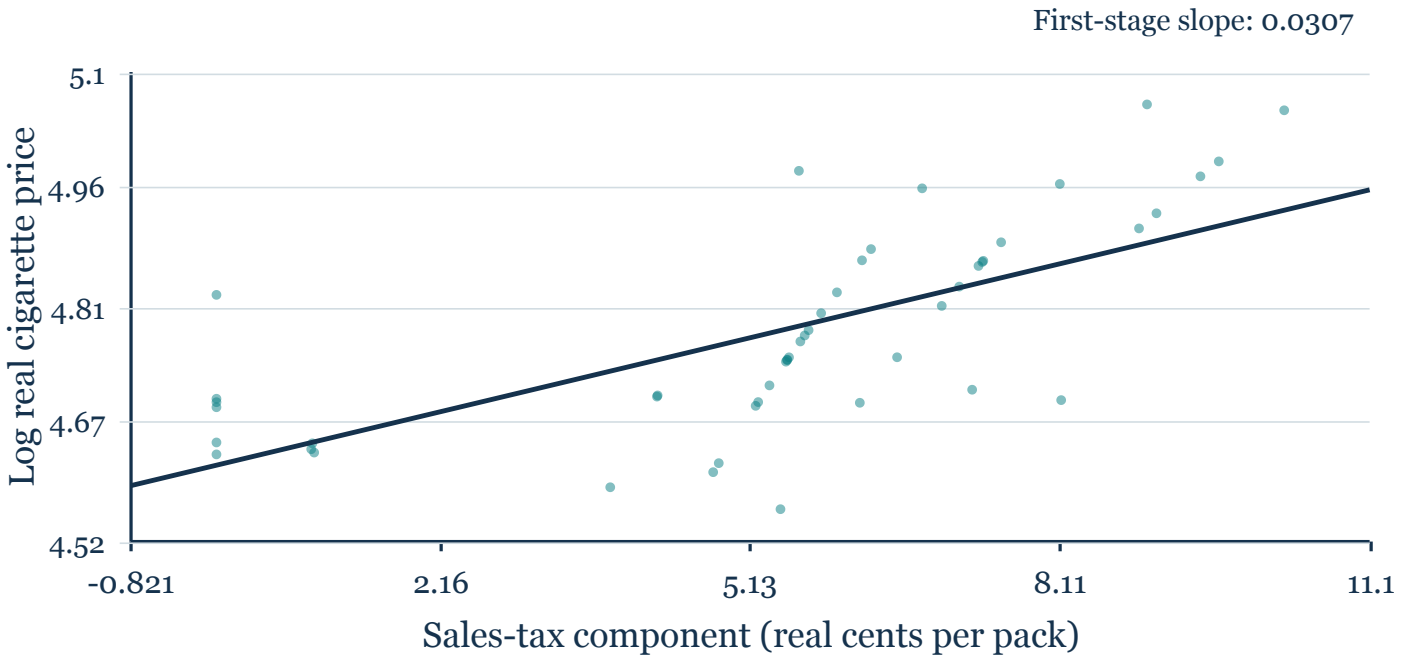
Do higher cigarette prices reduce sales?

Price is chosen in a market, so its variation may be endogenous.



Actual 1995 state price and tax data. The first stage describes relevance; it cannot establish the exclusion restriction.

- Question: isolate a demand response from other state differences.
- IV: instrumental variables; the proposed source of price variation needs its own argument.

Define the demand equation and its units

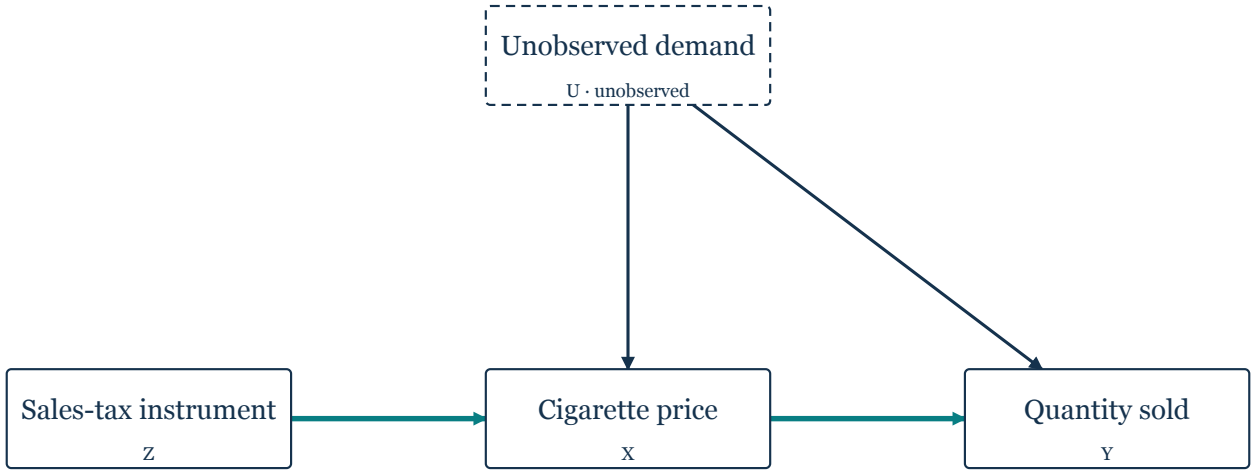
A log price slope is an elasticity under the model.

> $$\log Q_i = \alpha + \beta \log P_i + u_i$$

> Source deflator has base one: real cents per pack equal nominal cents divided by the recorded price index.

Relevance is the first instrument requirement

The instrument must move the endogenous regressor.



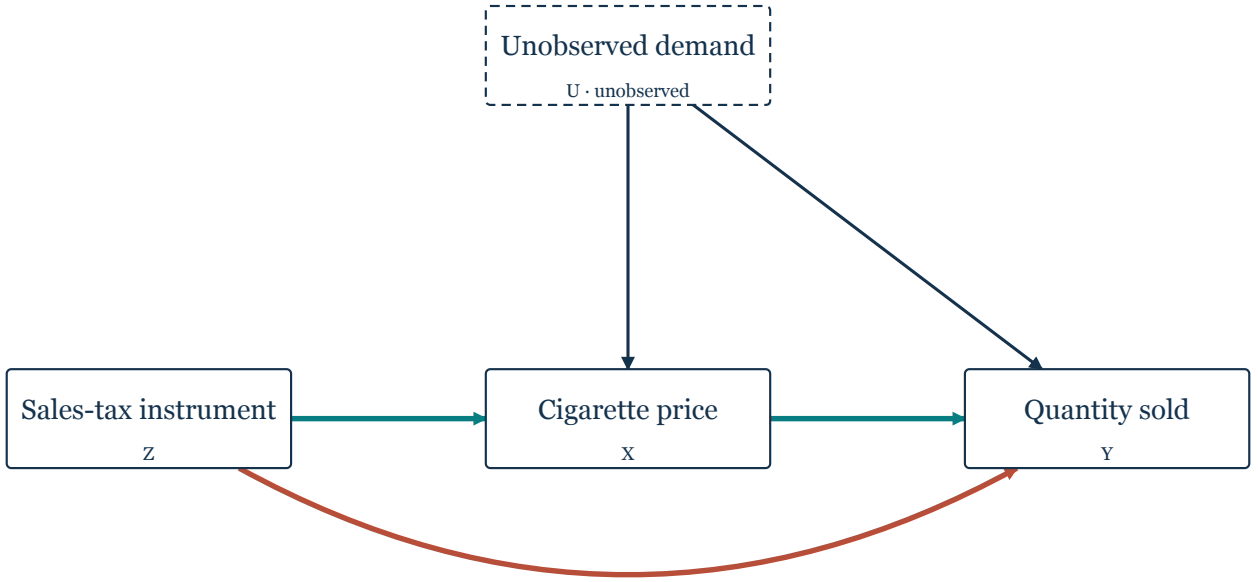
1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- First stage: $X = \pi_0 + \pi_1 Z + v$
- Relevance: $\pi_1 \neq 0$
- Z: real cents per pack; $X = \log P$

Exogeneity and exclusion justify the comparison

A strong first stage does not establish either condition.



- Exogeneity: no instrument relationship with omitted structural demand causes.
- Exclusion: no causal instrument-to-quantity channel bypassing price.
- Required IV moment: $\text{Cov}(Z, u) = 0$

Derive the simple IV ratio

Compare how the instrument moves the outcome and the regressor.

> First stage: $X = \pi_0 + \pi_1 Z + v$

> Reduced form: $Y = c + \rho Z + \eta, \quad \rho = \beta \pi_1$

> $\widehat{\beta}_{IV} = \hat{\rho} / \hat{\pi}_1$

Wald ratio

Try the comparison before opening the explanation.

Practice

A proposed instrument increases treatment by 0.30 units; its reduced-form outcome coefficient is −0.09. Compute the simple IV ratio and name the conditions behind a causal interpretation.

Wald ratio: explanation

Follow the calculation and interpret its units.

> The ratio is −0.09 divided by 0.30, or −0.30 outcome units per treatment unit. Under the constant-slope structural model, relevance, instrument exogeneity and exclusion justify the causal interpretation. The reduced-form and first-stage regressions must use compatible controls and observations. A strong first stage alone does not establish instrument validity.

Inspect the OLS demand comparison first

It is a useful benchmark, not the target by default.

```
. regress logq logp, vce(robust)

------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        logp |  -1.213057   .1945896    -6.23   0.000    -1.604746   -.8213685
       _cons |   10.33892   .9348202    11.06   0.000     8.457229    12.22062
------------------------------------------------------------------------------
```

Read the actual first stage

State the instrument units as recorded in the data.

```
. regress logp salestax, vce(robust)

------------------------------------------------------------------------------
             |               Robust
        logp | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    salestax |   .0307289   .0048354     6.35   0.000     .0209956    .0404621
       _cons |   4.616546   .0289177   159.64   0.000     4.558338    4.674755
------------------------------------------------------------------------------
```

Read the reduced form using the same controls and sample

Its scale is quantity change per instrument unit.

```
. regress logq salestax, vce(robust)

------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
    salestax |  -.0332974   .0102541    -3.25   0.002    -.0539378    -.012657
       _cons |   4.717449   .0650586    72.51   0.000     4.586493    4.848405
------------------------------------------------------------------------------
```

What the two stages actually fit

Use the instrument-predicted part of price, with a dedicated IV variance calculation.

> C: included exogenous controls; γC: their coefficient-weighted sum.

> First stage: $\widehat{X} = \widehat{E}_{linear}(X \mid Z, C)$

> Structural residual: $\widehat{u} = Y - \widehat{\alpha} - \widehat{\beta} X - \widehat{\gamma} C$

> 2SLS uses instrument-predicted variation; IV inference uses the structural model.

Use the dedicated IV estimator for uncertainty

Manual fitted-regressor OLS is not enough for valid IV SEs.

```
ivregress 2sls logq (logp = salestax), vce(robust)
```

Two stages and uncertainty

Try the comparison before opening the explanation.

Practice

Why not report ordinary OLS standard errors from regressing y on first-stage fitted x?

Two stages and uncertainty: explanation

Follow the calculation and interpret its units.

> The second-stage fitted regressor is generated, and the relevant structural residual and variance calculation differ from the naive second-stage OLS calculation. Use a proper IV estimator such as ivregress 2sls with a justified covariance option.

Compare the actual IV coefficient with its ratio

The do-file verifies equality before rounding.

```
. ivregress 2sls logq (logp=salestax), vce(robust)

------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       logp |  -1.083587   .3122036    -3.47   0.001    -1.695495    -.471679
       _cons |   9.719877   1.496143     6.50   0.000      6.78749    12.65226
------------------------------------------------------------------------------
```

Included exogenous controls belong in the IV specification

They help define the same conditional comparison in both stages.

```
. ivregress 2sls logq logincome (logp=salestax), vce(robust)

------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        logp |  -1.143375   .3604805    -3.17   0.002    -1.849904   -.4368463
   logincome |   .2145153   .3018477     0.71   0.477    -.3770953    .8061258
       _cons |   9.430658   1.219402     7.73   0.000     7.040675    11.82064
------------------------------------------------------------------------------
```

Hold income fixed in both supporting IV regressions

Compatible controls and observations make the controlled ratio meaningful.

```
. ivregress 2sls logq logincome (logp=salestax), vce(robust)

------------------------------------------------------------------------------
             |               Robust
        logq | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
        logp |  -1.143375   .3604805    -3.17   0.002    -1.849904   -.4368463
   logincome |   .2145153   .3018477     0.71   0.477    -.3770953    .8061258
       _cons |   9.430658   1.219402     7.73   0.000     7.040675    11.82064
------------------------------------------------------------------------------
```

Actual StataMP output; reproduce with iv_extension.do in the Stata download. Coefficient table shown; the download retains the full model output.

- Use the same exogenous income control and 48 states in every supporting equation.
- Controlled ratio: $\widehat{\beta} = \hat{r}_Z / \hat{\pi}_Z$.
- Read the dedicated IV result for its uncertainty.

School availability illustrates the exclusion question

A plausible first stage can still leave another route to the outcome.

> Suppose nearby single-sex school availability predicts attendance at that school type, and the outcome is a student's score.

> Residential choice, local resources or school competition may create other relationships.

Unexpected first-stage sign

Try the comparison before opening the explanation.

Practice

School availability predicts less attendance at the intended school type. Is a negative first-stage coefficient alone proof that IV is invalid?

Unexpected first-stage sign: explanation

Follow the calculation and interpret its units.

> No. Relevance requires a nonzero first stage, not a prescribed positive sign. An unexpected sign should prompt checks of coding, selection, theory and strength. It does not by itself prove or disprove exclusion or exogeneity.

Whose treatment changes when encouragement changes?

A binary encouragement example introduces the treatment-response groups.

> Compliers: attend only if offered. Always-takers/never-takers: unchanged attendance.

> Monotonicity: the offer does not reduce any individual's attendance.

> Constructed ratio: $0.90/(0.60 - 0.30) = 3$ score points.

With heterogeneous effects, say whose response IV identifies

A constant coefficient can hide a local treatment comparison.

> LATE: average treatment effect for compliers under the binary-IV assumptions.

> An IV population is determined by the instrument and the maintained model.

Weak instruments make the ratio fragile

A near-zero first stage magnifies uncertainty.

```
. estat firststage

  First-stage regression summary statistics
  ------------------------------------------------------------------------------
               |             Adjusted      Partial       Robust
      Variable |   R-sq.       R-sq.        R-sq.       F(1,45)   Prob > F
  -------------+----------------------------------------------------------------
          logp |  0.6389      0.6228       0.5009       44.7305    0.0000
  ------------------------------------------------------------------------------
```

> Weak first stage → denominator uncertainty can dominate the ratio.

> Strength and validity are different requirements.

A weak-instrument-robust test asks a stated coefficient question

Protecting inference from a weak first stage does not validate the instrument.

```
. estat weakrobust, betanull(-1)

Test robust to weak instruments
Model VCE: Robust

 ( 1)  logp = -1

Anderson–Rubin chi2(1) =   0.16
           Prob > chi2 = 0.6914

Note: Anderson–Rubin test reported by
      default because model is just
      identified.
```

Actual StataMP output; reproduce with iv_extension.do in the Stata download.

- For a proposed $\beta_0 = -1$, consider $\log Q - \beta_0 \log P = \log Q + \log P$.
- Anderson–Rubin: test the instrument restriction under the proposed coefficient.
- Validity still requires exogeneity, exclusion and justified sampling/inference conditions.

A larger instrument set adds assumptions as well as information

More instruments are not automatically a better design.

> Overidentified: more excluded instruments than endogenous regressors.

> A compatibility test cannot certify all substantive instrument assumptions.

Instrument scaling does not change the simple ratio

Consistent numerator and denominator transformations cancel.

> $\dfrac{\hat{\rho}/c}{\hat{\pi}_1/c} = \dfrac{\hat{\rho}}{\hat{\pi}_1}$ for nonzero c

Write the IV argument as a chain of evidence

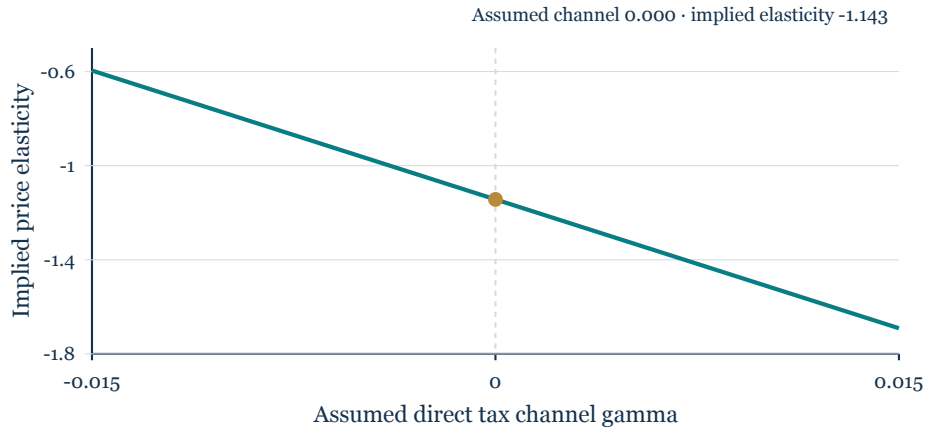
Separate relevance, exclusion, estimation and inference.

> Begin with the endogenous comparison and the omitted forces that threaten OLS.

> This sequence helps a policy reader assess the strategy rather than treat 2SLS as a label for causality.

How sensitive is IV to an excluded tax channel?

Treat an assumed exclusion violation as a sensitivity scenario, not an estimate.



Supporting regression slopes are actual income-controlled estimates. The extra direct tax channel is assumed for sensitivity, not estimated.

> Extended reduced form: $r = \beta\pi + \gamma$.
>
> Assumption-based sensitivity: $\beta(\gamma) = (r - \gamma)/\pi$.
>
> The direct-channel value is assumed; the supporting regression slopes are actual estimates.

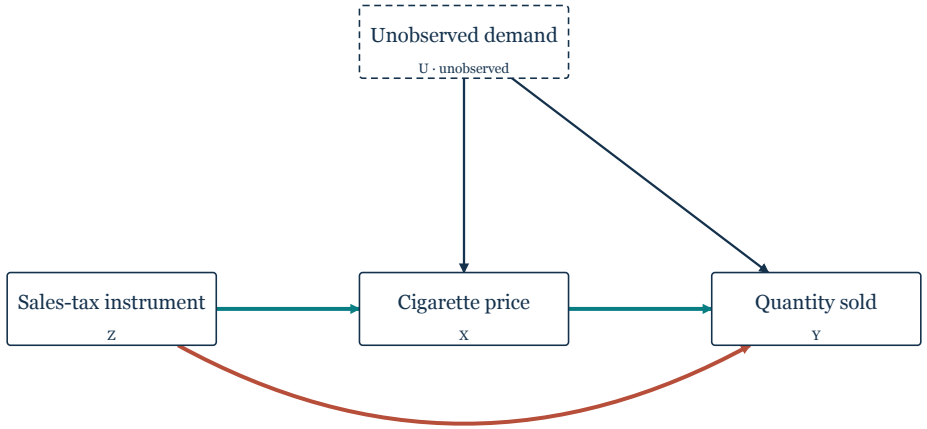
Practice

Is the direct-channel value gamma estimated by this exercise?

Explanation: No. It is an explicitly assumed sensitivity value.

DAG practice: challenge exclusion with one extra arrow

Trace the routes before choosing a control.



Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

The tax instrument predicts cigarette price. Add a direct tax-to-quantity arrow that bypasses price. Could the first stage stay strong? Why would dividing the reduced form by the first stage then fail to isolate the intended price elasticity?

DAG practice: challenge exclusion with one extra arrow: explanation

Explain what conditioning changes in this stated graph.

> The first stage can remain strong because the tax-to-price arrow has not disappeared. Relevance does not rule out a second route from the instrument to quantity that bypasses price. The reduced-form instrument association would combine the price channel and that additional direct channel. Dividing by the first stage no longer isolates the price response under the original exclusion argument. Use the graph to name a concrete institutional threat and assess it separately from the numerical strength of the first stage.

Reconstruct IV and assess the instrument separately

Reproduce the evidence, then explain its economic meaning.

Practice

Add log real income to the first stage, reduced form and IV equation, using the same 48 states. Verify the controlled ratio and compare the elasticity with the original no-income result. Keep your commands, log, output and interpretation.

Reconstruct IV and assess the instrument separately: explanation

A reproducible result needs both a calculation and an interpretation.

> Original IV elasticity: −1.083587.

> Income-controlled IV: −1.143375, robust SE 0.360481.

> Both stages need the same controls and states; exclusion remains a substantive argument.

Next, change the outcome from a level to a probability

A binary response requires a different interpretation of the fitted function.

> Next outcome: a zero-or-one event.

> Compare models on the probability scale after defining their functions.